

SIH PS26168

INTELLIGENT DEAD RECKONING

Project Documentation - Current Architecture, Evidence, Requirements and Status

Problem Statement	SIH PS26168 - Intelligent Dead Reckoning
Project snapshot	GitHub main at commit da84645 - "Added Live Map"
Repository	SaiChayank/intelligent-dead-reckoning
Primary application	Android / Kotlin / Jetpack Compose
Current evidence date	25 September 2026
Document purpose	Consolidated source of truth for current implemented, verified, experimental and planned capabilities

Status boundary: Offline map rendering, acquisition, recording, local export and Kotlin replay are implemented. A production navigation engine, corrected live INS, AI correction, EKF/UKF fusion, map matching, routing and turn-by-turn navigation are not yet implemented.

Prepared from the current GitHub repository, its verification documents, the frozen project decisions established during development, and the structure/traceability discipline used in the PS26145 reference document set.

Document Map

This document is intentionally broader than a conventional README. It combines the requirements baseline, current repository architecture, empirical data findings, implemented mobile subsystems, verification evidence, scientific limitations, optional map enhancement status, open gaps and traceability needed before the Master Blueprint and Implementation Plan are generated.

- 1. Source hierarchy and classification model
- 2. Problem statement and objectives
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1. Source Hierarchy and Classification Model

1.1 Source priority

When sources conflict, this documentation uses the following priority. This prevents historical planning documents from overriding later verified implementation evidence.

1. Official SIH PS26168 requirements and user-approved frozen decisions.
2. Current GitHub source at main / commit da84645 and the behavior represented by that source.
3. Executed verification records in mobile/VERIFICATION.md, RECORDING.md, EXPORT.md, REPLAY.md, OFFLINE_MAP.md and MAP_DEVICE_VERIFICATION.md.
4. Empirical research reports under reports/, especially synchronization, body-frame, frame-export and Phase 1 INS evidence.
5. Older planning PDFs and historical README statements. These remain useful context but are superseded where the implementation or newer evidence says otherwise.

1.2 Classification labels

Label	Meaning
OFFICIAL / FROZEN REQUIREMENT	Problem-statement requirement or requirement explicitly frozen for this project.
DERIVED REQUIREMENT	Necessary engineering consequence of the official goal, but not presented as verbatim SIH text.
TEAM DECISION	Architecture or implementation choice adopted by the project.
OPTIONAL ENHANCEMENT	Useful SIH/demo feature that is not required to satisfy the core problem statement.
IMPLEMENTED	Present in the current repository.
VERIFIED	Implemented and supported by executed automated and/or physical evidence.
EXPERIMENTAL	Research code/evidence exists, but it is not approved for production integration.
PLANNED	Accepted future component not yet implemented.
BLOCKED	Future component whose technically credible implementation depends on an unresolved prerequisite.

1.3 Important repository-status note

The root README currently contains stale statements that recording, replay and maps are still missing. The mobile subsystem documentation and current source are newer and show that local recording, export, Kotlin replay and offline map rendering have since been implemented. Therefore, the root README must not be used by itself as the current project status source.

2. Problem Statement and Objectives

2.1 Core problem

The project addresses SIH PS26168 - Intelligent Dead Reckoning for vehicle navigation when GNSS/GPS becomes unavailable, degraded or unreliable. The system is intended to preserve a useful estimate of position and motion from inertial sensing, then recover cleanly when GNSS returns. The project deliberately avoids dependence on vehicle OBD/CAN signals so that the main prototype can operate from phone/IMU measurements and later support a higher-rate edge target.

2.2 Primary objective

Build an on-device navigation pipeline capable of continuing vehicle position estimation through GNSS outages using calibrated inertial measurements, learned error correction and sensor fusion, while exposing uncertainty and avoiding unsupported claims when the navigation stack is not yet validated.

2.3 Frozen performance and deployment targets

Target	Classification	Current status
Positional drift during GNSS outage: target <10% of travelled distance	OFFICIAL / FROZEN TARGET	Not yet demonstrated by the current live navigation stack; historical raw INS is far outside target.
Android application as main prototype	TEAM DECISION / FROZEN	Implemented foundation, acquisition, recording, export, replay and offline map.
On-device operation without mandatory cloud inference	FROZEN REQUIREMENT	Current Android features operate locally; map pack is bundled and app has no INTERNET permission.
No OBD/CAN dependency for the primary solution	FROZEN REQUIREMENT	Preserved.
Approx. 10 Hz mobile navigation output target	FROZEN DESIGN TARGET	Navigation engine not yet implemented.
Higher-rate edge target around 200 Hz with FOG-class IMU	FROZEN DESIGN TARGET	Edge engine not yet implemented.

2.4 Success definition

A credible final system must do more than animate a map. It must demonstrate traceable sensor inputs, causal calibration, physically valid mechanization, AI correction that improves held-out error, GNSS-quality aware fusion, explicit confidence, bounded computation, safe GNSS-to-DR and DR-to-GNSS transitions, and repeatable testing in real and replayed outage scenarios.

3. Scope, Users and Operating Scenarios

3.1 Target users

User	Need
Vehicle driver / navigator	Continuous, understandable vehicle position during tunnels, parking structures and GNSS-degraded areas.
SIH evaluator / demonstrator	Visible evidence that the system continues using inertial/dead-reckoning output rather than simply freezing GPS.

User	Need
Developer / researcher	Replayable typed measurements, deterministic contracts, diagnostics, evaluation metrics and offline model development.
Future edge integrator	A navigation core that can be reused with higher-rate IMUs without coupling to Compose or phone UI.

3.2 Primary scenarios

- Normal outdoor operation with usable GNSS.
- GNSS degradation in urban canyons or multipath.
- GNSS denial in tunnels or covered parking structures.
- Continued dead-reckoning propagation during the outage.
- GNSS reacquisition with controlled convergence rather than a visible position jump.
- Offline replay of recorded phone sessions for reproducible development and evaluation.
- Offline map visualization without mandatory internet access.
- Future higher-rate edge execution with a better IMU.

3.3 Out of current implemented scope

- Production-grade live INS/EKF/AI navigation on the phone.
- Validated phone-to-vehicle mounting calibration for live navigation.
- Offline routing/re-routing and turn-by-turn guidance.
- Road map matching/HMM/Viterbi production logic.
- Background navigation service.
- Cloud inference or cloud map dependency.
- Automatic claims that the new phone recorder validates historical IO-VNBD export semantics.

4. Functional and Non-Functional Requirements

4.1 Functional requirements

Requirement	Classification	Current status
Acquire accelerometer, gyroscope, magnetometer and optional gravity from Android	FROZEN REQUIREMENT	VERIFIED
Acquire permitted GNSS/network location with nullable speed/bearing/accuracy fields	FROZEN REQUIREMENT	VERIFIED
Preserve monotonic event and receipt timestamps	FROZEN REQUIREMENT	VERIFIED

Requirement	Classification	Current status
Record typed measurement streams locally and recover interrupted sessions	DERIVED REQUIREMENT	VERIFIED
Export a session explicitly without cloud upload	DERIVED REQUIREMENT	VERIFIED
Replay saved sessions with replay source identity	DERIVED REQUIREMENT	VERIFIED
Calibrate device/vehicle relationship causally	CORE REQUIREMENT	PLANNED / BLOCKED by unresolved frame evidence for legacy dataset; live-phone calibration still future
Run inertial mechanization from calibrated IMU	CORE REQUIREMENT	EXPERIMENTAL in Python only; not approved for live use
Apply AI-based correction to reduce inertial error	CORE REQUIREMENT	PLANNED
Fuse GNSS and inertial navigation through EKF/UKF or approved fusion architecture	CORE REQUIREMENT	PLANNED
Apply vehicle constraints	DERIVED REQUIREMENT	PLANNED
Expose position, velocity, heading, confidence and navigation status	CORE REQUIREMENT	Contract exists; engine producer not implemented
Render offline map and display navigation output	OPTIONAL ENHANCEMENT	Offline renderer VERIFIED; real navigation connection PLANNED
Offline map matching / routing / turn-by-turn	OPTIONAL / HIGH-VALUE	Not implemented

4.2 Non-functional requirements

Requirement	Why it matters	Status
On-device / offline-first	GNSS-denied environments may also have poor connectivity; SIH prototype must not depend on cloud availability.	Current acquisition, recording, replay and map render locally.
Bounded queues and memory	Long recordings and high-rate sensors must not grow memory with trip duration.	Verified for acquisition and recorder; replay uses bounded policies.

Requirement	Why it matters	Status
Causal processing	No future information may be used in deployable navigation or training features presented as deployable.	Contract and methodology enforce this; historical evaluation baseline explicitly labeled when it uses VBOX/future reference.
Exact timestamp preservation	Sensor fusion and replay depend on reliable time.	Verified in contracts/recording/replay.
No silent unit/frame guessing	The legacy dataset contains ambiguous/misleading labels.	Explicit project rule.
Graceful lifecycle behavior	App backgrounding must not silently keep collecting/recording.	Verified foreground-only behavior.
Explainable status and uncertainty	Users/judges must distinguish GNSS, degraded, synthetic and future DR/fused states.	Partially available; explicit GNSS/DR/FUSED mode field is still missing from v1 navigation contract.

5. Current Project Architecture

5.1 Current implemented architecture

Android sensors / location

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AndroidAcquisition

| typed contracts/v1 Record

+-----> UI diagnostics

|

+-----> LocalRecorder

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+--> metadata.json

+--> measurements.jsonl

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+--> local ZIP export

+--> Kotlin replay

Separate optional map path:

SyntheticMapDemo -> MapDemoController -> NavigationPresentation

-> MapPresentation -> MapOverlay -> MapLibreRenderer

-> bundled Hyderabad MBTiles

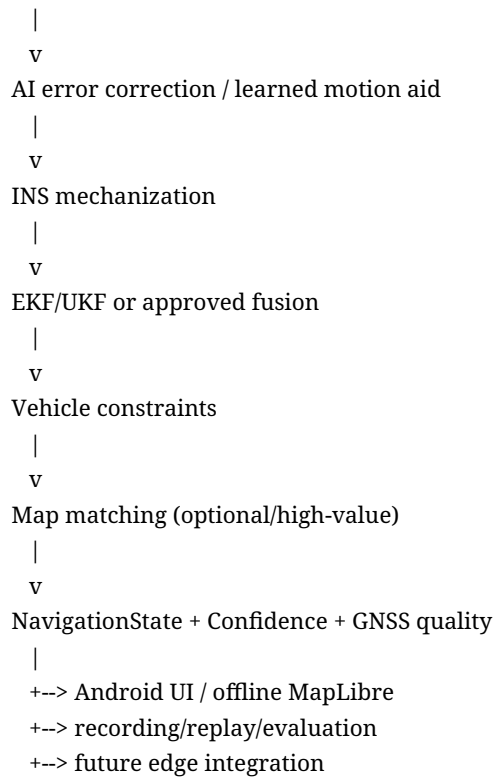
5.2 Intended final navigation architecture

IMU / GNSS

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Calibration and frame resolution



5.3 Architectural boundary that must remain frozen

The map is a consumer of navigation output. It must never become a second localization engine. Likewise, recording is downstream of acquisition and may not insert blocking file I/O into sensor or location callbacks. The shared contracts may not import Android UI or map SDK classes.

6. Repository and Module Structure

Path	Role	Status
contracts/v1/	Canonical cross-language navigation/measurement contract and strict Python/Kotlin codecs.	IMPLEMENTED / VERIFIED
contracts/recording/v1/	Versioned recording metadata contract and Kotlin/Python codec.	IMPLEMENTED / VERIFIED
mobile/	Main Android application, acquisition, recording, sessions, replay, offline map, UI and device evidence.	IMPLEMENTED / VERIFIED in bounded scopes
training/	Python dataset analysis, synchronization, frame diagnostics and experimental INS research.	IMPLEMENTED / EXPERIMENTAL
reports/	Empirical evidence, schema reports, frame studies and historical Phase 1 metrics.	IMPLEMENTED evidence store

Path	Role	Status
tests/	Python foundation/contract/frame/data tests.	IMPLEMENTED
docs/	Older planning/documentation PDFs.	Historical; requires refresh to current state
core/	Planned shared navigation core described in architecture, but not present as a completed implementation.	PLANNED
edge/	Planned higher-rate deployment target.	PLANNED
models/	Planned trained/exported model artifacts.	PLANNED / absent from current tracked tree

7. Data Foundation and IO-VNBD Evidence

7.1 Primary research dataset

IO-VNBD is the primary offline dataset used for research and early navigation diagnostics. Raw data is intentionally excluded from Git and is expected under data/raw/iovnbd. The repository preserves derived schema, synchronization, frame and integrity evidence rather than committing the raw corpus.

7.2 Important dataset findings

Finding	Implication
Representative synchronized smartphone data is nominally about 10 Hz.	Do not assume every repeated GPS value is a fresh hardware fix.
Smartphone GPS SPEED column is labelled Kmh, but empirical comparison supports m/s.	Never trust the source header blindly; conversion to km/h must be explicit.
Sequence M has two DATE gaps of approximately 1.158 s and 1.365 s over the full trace.	Row count equality does not prove perfect timing synchronization.
Manual synchronization exists, but exact time alignment is not established for all sequences.	Splits/evaluation must preserve synchronization uncertainty.
Categorized gyro Yaw/Pitch/Roll correspond to X/Y/Z in alternate copies.	This proves copy/header correspondence, not vehicle yaw/pitch/roll axes.
Acceleration/gravity exports appear world-like while other channels retain device-oriented semantics.	A single fixed raw-phone-to-vehicle rotation is not currently defensible for the legacy export.

Finding	Implication
Three unfiltered frame hypotheses passed 0/127 eligible windows; filtered reconstruction passed 1/127.	No universal calibration/frame mapping is approved.

7.3 Historical INS evidence

The historical 60-second M baseline is reproducible but not physically credible enough to serve as an accepted dead-reckoning baseline. It uses a nominal row clock and evaluation-only reference assistance. Reported results include approximately 429.788 m of reference travel, 1,644.650 m final position error, 382.67% drift relative to distance and about 126.27 km/h velocity RMSE. This is evidence that the existing mechanization assumptions are wrong or incomplete, not evidence that ordinary inertial drift is simply large.

7.4 Data-governance rules

- Do not modify data/raw.
- Do not silently relabel units or coordinate frames.
- Do not train on evaluation/test sequences or fit transforms using held-out future reference.
- Do not claim a phone recording validates historical IO-VNBD exporter semantics.
- Preserve scenario/driver separation in future train/validation/test splits.
- Record all dataset, feature, calibration and model versions needed to reproduce a result.

8. Navigation Exchange and Recording Contracts

8.1 Measurement/navigation contract v1

contracts/v1 defines strict UTF-8 JSON/JSONL typed records shared between Python and Kotlin. Event IDs and monotonic timestamps are decimal strings on the wire and decode to integer types without conversion through floating point. The contract rejects unknown keys, invalid enums, non-finite values, malformed JSON, duplicate event IDs and session/source mismatches.

8.2 Core payloads

Payload	Purpose
imu	Accelerometer, gyroscope, gravity or magnetometer sample with frame, unit and accuracy.
gnss	WGS84 location fix with nullable altitude, speed, bearing, accuracy, satellites, provider and UTC time.
calibration	Future causal calibration result including device-to-vehicle quaternion and biases.
navigation	Navigation status, mode, WGS84 origin, ENU position/velocity, vehicle attitude, heading and calibration reference.

Payload	Purpose
gnss_quality	GNSS quality state, fix age, satellites and reason codes.
confidence	Calibrated/unvalidated confidence plus horizontal 95% accuracy and speed uncertainty.
diagnostic	Stable machine-readable diagnostic code and dropped-count evidence.

8.3 Known contract gap for final live map

NavigationStatus describes uninitialized/calibrating/tracking/degraded/failed, but v1 does not contain an explicit current localization-mode field such as GNSS / DEAD_RECKONING / FUSED. The current map documentation correctly flags this. A future versioned contract decision is likely required before the final UI can label those modes without inference or overloading unrelated fields.

8.4 Recording contract

The recording contract stores session metadata in metadata.json and typed measurement records in measurements.jsonl. It preserves original acquisition source identity, timestamps, sensor metadata, permission/configuration state, completion/recovery state and finalized counts. Recovery is intentionally narrow: only a provably truncated trailing JSON fragment may be trimmed; complete interior corruption is rejected rather than silently skipped.

9. Android Acquisition Subsystem

9.1 Implemented behavior

- Foreground-only lifecycle ownership.
- Accelerometer and gyroscope requested near 100 Hz; gravity and magnetometer near 50 Hz where available.
- Location/GNSS permission states handled explicitly: not requested, denied, approximate, precise and revoked.
- Nullable location fields are preserved; absence is not converted to zero.
- Event timestamp and callback receipt timestamp are separate monotonic nanosecond fields.
- Bounded acquisition queue and explicit late/duplicate/drop/invalid diagnostics.
- Real and simulation sources are explicitly separated.
- Backgrounding/source switching stops the active source and does not silently resume on return.

9.2 Physical evidence on OnePlus 12R

Channel	Observed
Accelerometer	bmi26x / Bosch, approximately 98.79-98.80 Hz during verification
Gyroscope	bmi26x / Bosch, approximately 98.79-98.81 Hz

Channel	Observed
Gravity	QTI, approximately 49.39-49.40 Hz
Magnetometer	Memsic, approximately 50.00 Hz
Precise GPS in verified outdoor run	approximately 0.997 Hz when available
30-minute stationary run	526,748 accepted, 2 delayed, 0 duplicates/drops/invalid, final queue 0/256, peak 18/256

9.3 What acquisition does not prove

Sensor acquisition verifies that the phone stream is real, typed and bounded. It does not validate mounting calibration, inertial mechanization, AI correction, GNSS fusion, map matching or final navigation accuracy.

10. Recording, Export and Replay Subsystem

10.1 Local recorder

LocalRecorder subscribes downstream to the typed acquisition stream. A capacity-512 recorder channel separates the producer from disk I/O. Overflow or membership failure stops recorder admission and fails the recording explicitly rather than blocking acquisition or losing data silently.

10.2 Storage and recovery

- Sessions use app-private no-backup storage.
- Initial and final metadata use atomic replacement where supported.
- JSONL measurements are streamed rather than accumulated for the whole trip.
- Interrupted/open sessions are scanned and reconciled on recovery.
- Complete corrupt interior records are never silently repaired.
- Process interruption and controlled storage failure have physical/device evidence.

10.3 Final monitored recording evidence

Metric	Evidence
Duration	603.757 s (about 10 minutes)
Records	179,358 total
Recorder dropped	0
Write errors	0
ID gaps	0
Acquisition queue high-water	53 / 256 during monitored run

Metric	Evidence
Accelerometer / gyro	59,645 each, about 98.79 Hz
Magnetometer	30,187, 50 Hz
Gravity	29,821, about 49.39 Hz

10.4 Local export

The current app provides explicit local export through Android ACTION_CREATE_DOCUMENT. The archive contains exactly metadata.json and measurements.jsonl as byte copies of the private originals. No storage permission, background service or automatic upload is required. Cloud/unknown document providers are rejected by the current prototype policy.

10.5 Kotlin replay

Saved sessions can be replayed read-only with start/pause/resume/stop. Replay keeps the original measurements/timestamps/order and changes only the emitted source identity: real -> replay_real and simulation -> replay_simulation. It does not register live sensors. Current replay is fixed at 1x, has no seek/import UI and caps a session at one million records.

10.6 Remaining cross-language gap

The repository contains Python recording-contract codecs and has used Python scripts to validate saved sessions during evidence collection, but it does not yet have a clean reusable Python session-reader/summary/interoperability package equivalent to the Kotlin SessionFiles/ReplayReader layer. That remains a bounded missing component.

11. Offline Map and Synthetic Navigation Presentation

11.1 Scope classification

The live-navigation map is an OPTIONAL ENHANCEMENT, not a core official requirement. The current repository implements offline map rendering and a synthetic navigation presentation specifically so that map/UI work can progress without pretending the real navigation engine exists.

11.2 Current technology stack

Component	Current choice
Renderer	MapLibre Android OpenGL 13.6.1
Map data	OpenStreetMap-derived data using OpenMapTiles schema / OpenFreeMap snapshot
Packaging	Bundled Hyderabad MBTiles plus local glyph PBFs and manifest checksums

Component	Current choice
Runtime connectivity	MapLibre connected mode disabled; app removes INTERNET / network / Wi-Fi permissions
Coverage	Central Hyderabad: 17.30-17.55 N, 78.35-78.60 E
Map asset	hyderabad.mbtiles about 22.5 MB, 247 tile coordinates in verified pack

11.3 Implemented presentation features

- Current/synthetic position marker and heading polygon.
- Travelled trail.
- Synthetic uncertainty radius.
- Three selectable synthetic trajectories: left curve, straight and right curve.
- 0.5x / 1x / 2x virtual demo playback.
- Automatic 10-20 s synthetic blackout plus manual blackout/recovery control.
- Comparison marker/trail and scripted convergence.
- Road-overlay visibility toggle, scenario path and outage segment overlay.
- Distance, outage duration/distance, heading and speed HUD.
- Pan, zoom, north-up, recenter and visible attribution.

11.4 Reference-video provenance

MAP_DEMO_FEATURES.md documents which presentation ideas were inspired by each of the four supplied YouTube references. The implementation intentionally does not copy video code/artwork or adopt unsupported benchmark claims. The red comparison trace is scripted illustrative drift, not measured INS/AI error; the synthetic recovery is not an EKF correction.

11.5 Device evidence

On 25 September 2026 the OnePlus 12R physical verification confirmed that roads, labels, buildings and water rendered from the offline pack, synthetic marker/heading/trail/uncertainty rendered, camera controls worked and attribution remained visible. The stage records 116 JVM tests passing and a final four-test direct device run. This verifies rendering and synthetic presentation only.

12. Navigation, INS, AI and Fusion Status

12.1 NavigationEngine

contracts/v1 defines a UI-independent NavigationEngine interface with initialize, acceptImu, acceptGnss, drain, stop and reset. No production implementation exists. This is the primary integration gap between the verified sensor/recording foundation and the final Intelligent Dead Reckoning objective.

12.2 Calibration and frame resolution

The final live system requires a causal, validated device-to-vehicle calibration. Legacy IO-VNBD exports cannot currently provide a universally approved raw-device frame mapping. The new Android live

stream is better defined (android_device frame), but the phone-to-vehicle mounting/calibration algorithm still needs to be implemented and physically verified.

12.3 Classical INS

training/ins_mechanization.py remains experimental research. Its historical result is not accepted as a deployable baseline because frame assumptions and initialization contaminate the result. Before AI training, the project needs synthetic mechanization tests plus a physically credible corrected baseline on well-defined device-frame data.

12.4 AI correction

No trained AI correction model is currently committed. The future model must have a clearly defined target (for example speed/velocity correction or error-state aid), causal features, scenario-separated splits, leakage prevention, reproducible training artifacts and an on-device export format such as ONNX/TFLite only after validated improvement over the classical baseline.

12.5 Fusion

EKF/UKF fusion is planned, not implemented. It must consume approved navigation/IMU states and GNSS quality, preserve covariance/uncertainty semantics, continue through outages and handle reacquisition without a visible jump. Display smoothing must not substitute for state-estimation correctness.

13. Map Rendering, Positioning, Map Matching, Routing and Turn-by-Turn

These capabilities are separate and must remain separate in architecture and claims.

Capability	Definition	Current project status
Map rendering	Draw geographic data and overlays.	IMPLEMENTED / VERIFIED offline for Hyderabad subset.
Positioning	Estimate vehicle pose from GNSS/inertial/fusion outputs.	Real navigation engine NOT IMPLEMENTED; synthetic demo only.
Map matching	Constrain an estimated trajectory to plausible road geometry.	NOT IMPLEMENTED.
Routing	Compute a path through a road graph between origin and destination.	NOT IMPLEMENTED; MBTiles are not a routing graph.
Turn-by-turn navigation	Track route progress, maneuvers, ETA and re-routing.	NOT IMPLEMENTED.

13.1 Future offline routing strategy

If routing is added, it will require a local routable road graph and an embedded routing engine or equivalent custom graph layer. MapLibre/MBTiles alone cannot compute routes. The routing graph

should be region-scoped, versioned and packaged separately from rendering tiles so storage and licensing remain manageable.

14. Security, Privacy and Offline Constraints

14.1 Current Android controls

- No INTERNET permission in the installed application; renderer-inherited network/Wi-Fi permissions are explicitly removed.
- No background service or background-location permission.
- App backups are disabled.
- Recordings live in app-private no-backup storage.
- Export occurs only after explicit user action.
- Export retains the private original.
- No mandatory cloud inference or map service.
- Offline map pack uses checksums and a fixed resource manifest.
- Local session IDs and filenames are validated before file access.
- ZIP export uses fixed canonical entry names and does not expose arbitrary internal paths.

14.2 Security limitations

The current hackathon app does not implement user accounts, encrypted-at-rest trip storage, enterprise key management or production app hardening. Those may be production extensions, but they should not be added in a way that compromises offline operation or the deterministic recording/evaluation workflow.

15. Verification Evidence

15.1 Verification layers

Layer	Evidence
Python foundation/data	Schema, synchronization, frame and contract tests plus empirical reports.
Android unit/JVM	Acquisition, contract, recorder, export/session, replay, map-presentation and offline-map tests.
Android instrumentation	UI/lifecycle, permissions, recording, storage-failure, export, replay and offline-map device tests.
Physical acceptance	OnePlus 12R acquisition, sensor axes/rates, lifecycle cleanup, long recording, recovery/export/replay and map rendering.
Scientific navigation	NOT YET ACCEPTED. Historical raw INS metrics remain experimental and poor.

15.2 Key verified checkpoints

Checkpoint	Result
Prompt 2 real acquisition	8/8 connected tests in original acceptance; 30-minute stationary run; sensor/GNSS lifecycle verified.
Recording contract	Python 9/9 plus Kotlin/JVM/lint checks at contract stage.
Prompt 3B recorder	Final ten-minute monitored physical run: 179,358 records, 0 recorder drops/write errors/ID gaps; process recovery and controlled storage failures verified.
Local export	Physical SAF/Downloads exports verified; large ZIP entries SHA-256 matched private originals.
Kotlin replay	96 JVM tests and 19 device instrumentation tests at replay checkpoint; actual completed and recovered real sessions replayed.
Offline map	116 JVM tests at latest map stage and final four-test physical map run; no internet permission.

15.3 CI status

There are currently no GitHub Actions workflow runs or commit status checks attached to the latest main commit. Verification is documented as local host/device execution. A later productionization phase should add CI for deterministic host tests while keeping device-only acceptance separate.

16. Current Status Matrix

Component	Status	Action / boundary
Dataset inventory/schema/sync audit	VERIFIED	Keep frozen; refresh provenance only intentionally.
Body-frame/export investigation	EXPERIMENTAL / UNRESOLVED	Do not use legacy export as approved raw device IMU.
Historical INS baseline	EXPERIMENTAL / FAILED GATE	Replace with corrected mechanization after calibration/frame prerequisites.
Cross-language measurement contract	VERIFIED	Freeze v1 unless a versioned change is approved.
Android real acquisition	VERIFIED	Freeze behavior.
Recording contract	VERIFIED	Freeze v1.
Foreground recorder	VERIFIED	Freeze after current evidence.

Component	Status	Action / boundary
Session UI + local export	VERIFIED	Treat as current functionality.
Kotlin replay	VERIFIED	Treat as current functionality.
Python reusable Prompt-3 session reader	NOT IMPLEMENTED	Next bounded interoperability task if needed.
Offline MapLibre renderer	VERIFIED	Optional enhancement; frozen rendering boundary.
Synthetic map demo	VERIFIED AS DEMO	Never label as measured DR/AI.
NavigationEngine implementation	NOT IMPLEMENTED	Primary next core milestone.
Live calibration	NOT IMPLEMENTED	Required before credible INS.
Corrected live INS	NOT IMPLEMENTED	Blocked by calibration/mechanization work.
AI correction	NOT IMPLEMENTED	Requires validated target/baseline/data split.
EKF/UKF fusion	NOT IMPLEMENTED	Requires approved navigation state and GNSS policy.
Vehicle constraints	NOT IMPLEMENTED	Add after core motion state works.
Map matching	NOT IMPLEMENTED	Optional/high-value after navigation state is credible.
Offline routing / rerouting	NOT IMPLEMENTED	Requires routing graph/engine.
Turn-by-turn	NOT IMPLEMENTED	Future enhancement.
Edge engine	NOT IMPLEMENTED	Future deployment target.

17. Risks, Limitations and Unresolved Decisions

17.1 High-risk technical issues

Risk	Impact	Required handling
Legacy dataset frame ambiguity	Can invalidate mechanization and AI targets.	Do not proceed by guessing. Use clearly defined live-device frame and controlled calibration data; retain legacy dataset only where semantics are defensible.

Risk	Impact	Required handling
No credible production INS baseline yet	AI improvement cannot be measured against a trustworthy baseline.	Build synthetic mechanization tests and a physically valid baseline before training correction models.
No explicit GNSS/DR/FUSED mode field in v1	Map/UI cannot reliably label localization mode.	Version contract intentionally when engine semantics are finalized.
Offline map is region-limited	Demo outside central Hyderabad has no tiles.	Keep region packaging modular; add selected regions only when needed.
Routing graph absent	Map cannot provide real route planning.	Separate routing data/engine from MBTiles.
Only one physical Android device is deeply verified	Device-specific behavior may differ.	Add at least one additional device/API level before production claims.
No GitHub CI	Regression checks depend on manual/local execution.	Add CI for Python/JVM/lint/build once current working baseline is frozen.
Root README stale	New contributors may misunderstand project maturity.	Regenerate root README/status after this documentation set is approved.

17.2 Claims that must not be made

- That the synthetic map demo is a functioning AI dead-reckoning engine.
- That the red comparison drift is measured INS/AI performance.
- That the <10% drift target has already been achieved.
- That MBTiles provide map matching or routing.
- That new phone recordings validate IO-VNBD exporter frame semantics.
- That historical 382% drift is a normal acceptable baseline.
- That replay or map rendering proves navigation accuracy.

18. Requirements Traceability

Requirement	Mapped component	Status	Evidence / gap
Continue navigation through GNSS outage	NavigationEngine -> calibration -> AI/INS -> fusion	PLANNED	No production engine yet
<10% positional drift target	Evaluation harness / outage metrics	NOT MET / NOT YET TESTED credibly	Historical INS far outside target
No OBD/CAN dependency	Android acquisition + future core	PRESERVED	Phone sensors/GNSS currently used
Offline/on-device operation	Android app + local storage + MapLibre pack	VERIFIED for implemented subsystems	AI/navigation not yet present

Requirement	Mapped component	Status	Evidence / gap
Real sensor acquisition	AndroidAcquisition	VERIFIED	OnePlus 12R evidence
Reproducible sessions	Recording contract + LocalRecorder	VERIFIED	Exact timestamps/source/counts
Local export	SessionFiles + SAF	VERIFIED	ZIP byte preservation
Replay	ReplayReader/Controller	VERIFIED	replay_real mapping verified
Map visualization	MapLibreRenderer	VERIFIED as optional enhancement	Offline Hyderabad subset
GNSS/DR/fused map continuity	Future engine -> NavigationPresentation	NOT IMPLEMENTED	Synthetic only today
Map matching	Future matcher	NOT IMPLEMENTED	No road snap
Offline routing/re-routing	Future local routing graph/engine	NOT IMPLEMENTED	No routing graph
Edge deployment	Future edge engine	NOT IMPLEMENTED	Architecture target only

19. Frozen Decision Register

The following decisions should be treated as frozen unless new technical evidence proves one invalid or the project owner explicitly approves a revision.

- Android/Kotlin/Jetpack Compose is the main prototype application.
- Python remains the offline research/training/evaluation environment.
- Raw IO-VNBD data is not committed and must not be modified.
- contracts/v1 is the current canonical typed measurement/navigation exchange format.
- Recording uses metadata.json + measurements.jsonl and preserves exact source timestamps.
- Acquisition and recording are foreground-only; no silent restart after backgrounding.
- Recorder disk I/O is downstream and must never block sensor/location callbacks.
- Replay uses replay_real / replay_simulation and must not masquerade as live.
- Offline map rendering is an optional consumer of NavigationState, not a localization algorithm.
- MapLibre + local OSM/OpenMapTiles-derived data is the current offline-map stack.
- The map demo remains explicitly synthetic until a real NavigationEngine exists.
- No unsupported unit/frame conversion may be silently introduced to make INS appear to work.
- AI/EKF/map matching are not to be implemented as cosmetic substitutes for unresolved calibration/mechanization correctness.
- Any future conflict with these baselines must be raised explicitly rather than silently redesigned.

20. Documentation Maintenance and Evidence Sources

20.1 Repository evidence used

- Current main commit: da84645fc39f07363a58d3cee035bd196186f11d - Added Live Map.
- contracts/v1/README.md and Kotlin/Python models/codecs.
- contracts/recording/v1/README.md and recording codecs/models.
- mobile/ACQUISITION.md, VERIFICATION.md, RECORDING.md, EXPORT.md, REPLAY.md.
- mobile/OFFLINE_MAP.md, MAP_DEMO_FEATURES.md, MAP_DEVICE_VERIFICATION.md.
- mobile/app source for acquisition, recording, sessions, replay, map and UI.
- reports/data_dictionary.md, data_schema.md, synchronization_analysis.md.
- reports/frame_resolution_followup.md and frame-export evidence.
- reports/phase1_metrics.md and foundation_readiness.md.
- training/ analysis and experimental mechanization modules.
- PS26145 reference-document set supplied by the project owner, used for structure, classification discipline, traceability and audit style - not for transferring technical conclusions from the cyber-security project.

20.2 Update policy

This document should be revised after each major frozen milestone. Implemented/verified claims must be backed by code plus executed evidence. Planning language should never be silently rewritten as completed functionality. Historical sections should be preserved or explicitly superseded so that SIH evaluators and developers can distinguish design intent from executed proof.

20.3 Next document in this set

The next file should be the Master Blueprint. It will consume this documentation as the current-state source of truth and define the final target architecture, component contracts, phase gates, integration order and end-state acceptance criteria without reclassifying synthetic/demo features as production navigation.

END OF CURRENT-STATE PROJECT DOCUMENTATION